

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8" />

  <meta name="viewport" content="width=device-width, initial-scale=1.0" />

  <title>Gesture-Based Authentication System</title>

  <style>

    body {

      margin: 0;

      font-family: Arial, sans-serif;

      background: #f4f8fb;

      color: #222;

    }


    .container {

      max-width: 900px;

      margin: 40px auto;

      padding: 20px;

    }


    .card {

      background: white;

      border-radius: 14px;

      box-shadow: 0 4px 12px rgba(0,0,0,0.1);

      padding: 25px;

      margin-bottom: 20px;

    }


    h1, h2 {

      text-align: center;

      color: #0b5ed7;
```

```
}
```

```
.row {  
  display: flex;  
  gap: 20px;  
  flex-wrap: wrap;  
}
```

```
.box {  
  flex: 1;  
  min-width: 250px;  
  background: #eef5ff;  
  border-radius: 10px;  
  padding: 20px;  
  text-align: center;  
}
```

```
label {  
  font-weight: bold;  
  display: block;  
  margin-bottom: 10px;  
}
```

```
select, button {  
  width: 100%;  
  padding: 12px;  
  border-radius: 8px;  
  border: 1px solid #ccc;  
  font-size: 16px;  
  margin-top: 10px;  
}
```

```
button {  
  background: #0b5ed7;  
  color: white;  
  border: none;  
  cursor: pointer;  
}
```

```
button:hover {  
  background: #094db1;  
}
```

```
.status {  
  font-size: 20px;  
  font-weight: bold;  
  margin-top: 15px;  
}
```

```
.success {  
  color: green;  
}
```

```
.fail {  
  color: red;  
}
```

```
.info {  
  font-size: 18px;  
  margin-top: 10px;  
  color: #444;  
}
```

```
.footer {  
  text-align: center;  
  margin-top: 20px;  
  color: #666;  
}
```

```
.flow {  
  line-height: 1.8;  
  font-size: 17px;  
  text-align: center;  
  color: #333;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="container">
```

```
<div class="card">
```

```
<h1>Gesture-Based Authentication System</h1>
```

```
<p style="text-align:center;">
```

```
  Smart Home and IoT Security using Gesture Recognition
```

```
</p>
```

```
</div>
```

```
<div class="card">
```

```
<h2>Project Workflow</h2>
```

```
<div class="flow">
```

```
  Camera Input → Hand Detection → Gesture Recognition → Authentication → IoT Action
```

```
</div>
```

```
</div>
```

```
<div class="card">

  <h2>Authentication Demo</h2>

  <div class="row">

    <div class="box">

      <label for="gesture">Select Detected Gesture</label>

      <select id="gesture">

        <option value="">-- Choose Gesture --</option>

        <option value="OPEN_PALM">OPEN PALM</option>

        <option value="FIST">FIST</option>

        <option value="ONE">ONE</option>

        <option value="TWO">TWO</option>

      </select>

      <button onclick="authenticateUser()">Authenticate</button>

    </div>


```

```
    <div class="box">

      <h3>System Output</h3>

      <div id="authStatus" class="status">Waiting...</div>

      <div id="iotAction" class="info">No Action</div>

    </div>

  </div>

</div>
```

```
<div class="card">

  <h2>Technical Features</h2>

  <ul>

    <li>Gesture Recognition</li>

    <li>Authentication Verification</li>

    <li>IoT Device Control</li>

    <li>Real-Time Processing</li>

    <li>Smart Home Security</li>


```

```
</ul>
```

```
</div>
```

```
<div class="footer">
```

```
    Project Demo UI
```

```
</div>
```

```
</div>
```

```
<script>
```

```
function authenticateUser() {
```

```
    const gesture = document.getElementById("gesture").value;
```

```
    const authStatus = document.getElementById("authStatus");
```

```
    const iotAction = document.getElementById("iotAction");
```

```
    if (gesture === "OPEN_PALM") {
```

```
        authStatus.innerHTML = "Authenticated";
```

```
        authStatus.className = "status success";
```

```
        iotAction.innerHTML = "Door Unlocked";
```

```
    } else if (gesture === "FIST") {
```

```
        authStatus.innerHTML = "Access Denied";
```

```
        authStatus.className = "status fail";
```

```
        iotAction.innerHTML = "Door Locked";
```

```
    } else if (gesture === "ONE") {
```

```
        authStatus.innerHTML = "Authenticated";
```

```
        authStatus.className = "status success";
```

```
        iotAction.innerHTML = "Light ON";
```

```
    } else if (gesture === "TWO") {
```

```
        authStatus.innerHTML = "Authenticated";
```

```
        authStatus.className = "status success";
```

```
        iotAction.innerHTML = "Fan ON";
```

```
    } else {
```

```
    authStatus.innerHTML = "Please Select a Gesture";  
    authStatus.className = "status fail";  
    iotAction.innerHTML = "No Action";  
  }  
}  
</script>  
</body>  
</html>
```